

電腦對局理論 Homework 2

蔡平樂

December 5, 2025

1 Implementation detail

The following is the design be used at the final submission of the MCTS agent.

1.1 Simulation and improvement

For each single MCTS iteration, the agent will perform 10 simulations, and use a PieceValue table to evaluate each move.

```
1: procedure MOVEEVALUATE(move)
2:    $s \leftarrow \text{move.from}$ 
3:    $e \leftarrow \text{move.to}$ 
4:   if  $e \neq \text{Null}$  then
5:      $\text{capture\_move\_value} \leftarrow \text{PieceValue}(e)$ 
6:     if  $s \neq \text{Cannon}$  then
7:        $\text{capture\_move\_value}+ = \frac{\text{PieceValue}(s)}{2}$ 
8:     return  $C \times \text{capture\_move\_value}$ 
9:   return  $\text{basic\_move\_value}$ 
```

basic_move_value and C are constants, which are

1.2 RAVE

The Rapid Action Value Estimation method only being used at the leaf. More precisely, I create two types of Nodes, Nodes and Nodes_AMAF, then update them respectively at a simulation

procedure UPDATENODE(node, sim_record)

1.3 Evaluation function design

2 Experiments